AI-PREDICTED – NOT AN OFFICIAL QUESTION PAPER (FOR REVISION & PRACTICE ONLY)

MAHARASHTRA STATE TECHNICAL UNIVERSITY (AUTONOMOUS)

BILINGUAL EXAMINATION PAPER / [illegible]

Database Management Systems & Information Architecture (CS-402)

Time: 3 Hours | **Total Marks / [illegible]: 100**

Q.1 **(English)**
Demonstrate the use of GROUP BY and HAVING clauses in SQL. Differentiate between WHERE and HAVING with queries.

([illegible] / [illegible])
Demonstrate the use of GROUP BY and HAVING clauses in SQL. Differentiate between WHERE and HAVING with queries. ([illegible].)

[20]

Q.2 **(English)**
Discuss ACID properties in transaction processing. Explain how each property is preserved by the DBMS recovery manager.

([illegible] / [illegible])
Discuss ACID properties in transaction processing. Explain how each property is preserved by the DBMS recovery manager. ([illegible].)

[20]

Q.3 **(English)**
Explain the Strict Two-Phase Locking (2PL) protocol. How does 2PL ensure serializability?

([illegible] / [illegible])
Explain the Strict Two-Phase Locking (2PL) protocol. How does 2PL ensure serializability? ([illegible].)

[20]

Q.4 **(English)**
Describe Deadlock handling mechanisms: Wait-Die scheme, Wound-Wait scheme, and Wait-For Graph (WFG) detection.

([illegible] / [illegible])
Describe Deadlock handling mechanisms: Wait-Die scheme, Wound-Wait scheme, and Wait-For Graph (WFG) detection. ([illegible].)

[20]

Q.5 **(English)**
Discuss mapping cardinalities (1:1, 1:N, M:N) and structural constraints in ER diagrams.

([illegible] / [illegible])
Discuss mapping cardinalities (1:1, 1:N, M:N) and structural constraints in ER diagrams. ([illegible].)

[20]